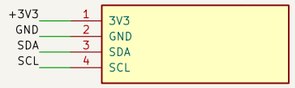
HastaDAQ

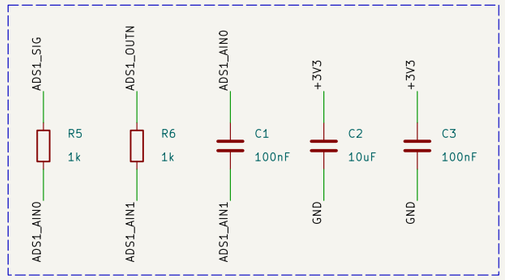
Net labels connect everything: same name = same net. +3V3 / GND are plain labels, not power symbols.

Pimoroni Pico Plus 2 (+ RM2 on SP/CE): same 40-pin header as Pico 2 W. Qwiic port = GP4/GP5 (I2C0), same nets as header pins 6/7 and 3V3 as pin 36. Each ADS122C04 board also has two Qwiic connectors in parallel with its VIN/GND/SDA/SCL pins. Crossed-out pins are intentionally unused.

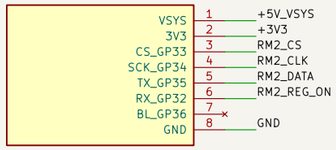
U2
Adafruit ICM20649 IMU (Qwiic, 0x68)



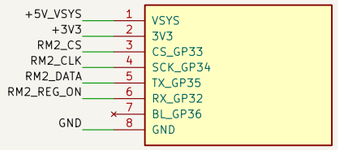
ADS1 input filter + supply decoupling (place at ADC)



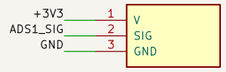
J4
Pico Plus 2 SP/CE connector (8-pin JST-SH)



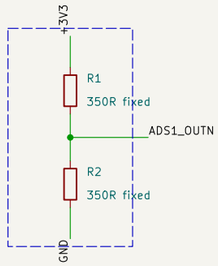
U5
Pimoroni RM2 WiFi/BT breakout (1:1 JST-SH cable)



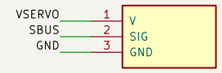
J1
HB1 half-bridge #1 (on beam)



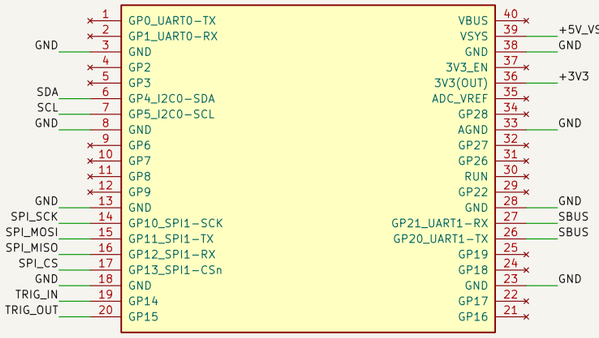
ADS1 completion resistors (fixed)



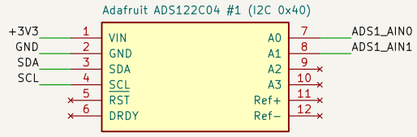
J5
SBUS out (V, SIG, GND)



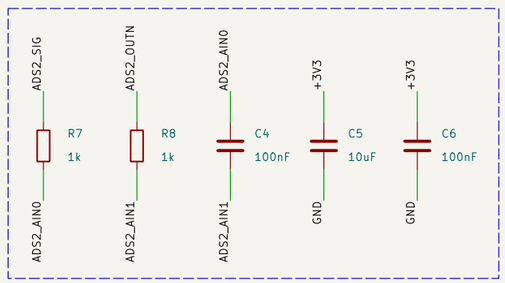
U1
Pimoroni Pico Plus 2 (header pins)



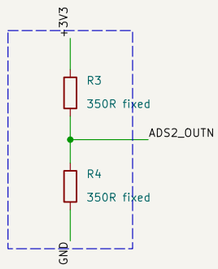
U3
Adafruit ADS122C04 #1 (I2C 0x40)



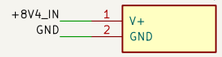
ADS2 input filter + supply decoupling (place at ADC)



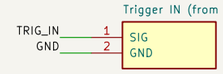
ADS2 completion resistors (fixed)



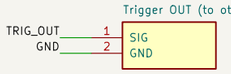
J6
Bench PSU in (8.4 V)



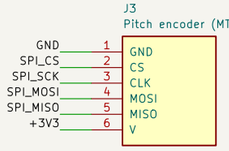
J7
Trigger IN (from other equip.)



J8
Trigger OUT (to other equip.)



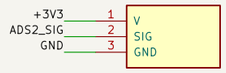
J3
Pitch encoder (MT6835)



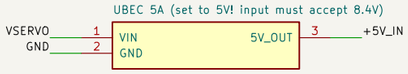
U4
Adafruit ADS122C04 #2 (addr jumper bridged)



J2
HB2 half-bridge #2 (on beam)

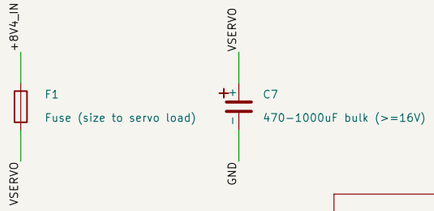


U6
UBEC 5A (set to 5V! input must accept 8.4V)



D1
Schottky (e.g. SS34)





NOTES
1. ONE common GND (Pico, sensors, servo PSU). SBUS needs a shared ground reference.
   RM2 WiFi breakout: SP/CE uses GP32-GP36 only, no overlap with Qwiic (GP4/5) or any other signal.
2. ADS2: bridge one address jumper on its back (A0/A1); resulting address unconfirmed -- firmware falls back to an I2C scan at boot.
3. Each half-bridge module (2 gauges) plugs into a 3-pin header: V, SIG (mid-point), GND. The two fixed 350R completion resistors sit next to the ADC; their mid-point goes to A1. Excitation is the ADC's own VIN, so VREF_SUPPLY is ratiometric -- no Ref+/Ref- wiring.
4. SBUS: GP20 (TX) and GP21 (RX) on one net (UART inverted in firmware).
5. POWER: 8.4 V bench PSU -> fuse -> VSERVO feeds the servos directly and the UBEC input. UBEC 5 V -> Schottky -> Pico VSYS. Pico VSYS max is 5.5 V (datasheet), so the UBEC MUST be set to 5 V, not 6 V. Confirm the UBEC's input range covers 8.4 V. USB (VBUS) is left for programming. Set a current limit on the bench PSU.
6. Trigger pins are 3.3 V logic only. Pin order on the headers: see each symbol.
7. Per ADC: 1k in each bridge-output leg + 100nF across them (~800 Hz corner), and 10uF + 100nF across the bridge supply. Place these at the ADC end. Use shielded twisted cable for the gauge wires, shield to GND at the ADC end only.

HastaArduino
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Title: HastaDAQ

Size: A3	Date:	**Rev: A**
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